

Using the FLIR Cameras

FLIR Camera Instructions and Troubleshooting

These instructions explain how to sign out, set up, use, and troubleshoot the FLIR ONE Pro infrared cameras. These cameras attach directly to a phone and allow users to take thermal images using the FLIR ONE app.

Most of the cameras use **USB-C**, but a few use **Lightning** connectors. Please make sure you sign out the correct camera type for your phone.

Accessing the Camera Guides

Before using the cameras, you can review the following resources:

FLIR ONE PRO User Guide:



Video "Basics of Infrared Thermography":



Video Example "Camera Inspection":



Signing Out a Camera

1. Go to the camera sign-out station.
2. Choose a camera that matches your phone type:
 - a. **USB-C** for most newer Android phones and some newer iPhones
- Note: **Read the camera to see if it says for iOS or for USB-C**
- **Lightning** for older iPhones
3. Record your group number, name, timeslot, phone connector type, and camera number on the sign-out sheet.
4. Before leaving the sign-out area, test the camera to confirm that it works.
5. Take a quick test photo to make sure the thermal image is visible and the camera is connected properly.

A photo of the sign-out station or an example of a properly completed sign-out row would be useful here.

Setting Up the FLIR Camera

1. Download the **FLIR One app** from the Apple App Store or Google Play Store.
2. Turn on the FLIR camera using the power switch on the device.
3. Connect the FLIR camera to your phone.
4. Open the FLIR ONE app.
5. Wait for the camera to connect and show a live thermal image.
6. If prompted, allow the app to access your camera, photos, and location.

If the app does not recognize the camera, disconnect the camera, close the app, reopen it, and reconnect the camera.

A screenshot of the FLIR ONE app home screen would be useful here.

Taking a Thermal Photo

1. Make sure the app is set to **Photo** mode.
2. Point the camera at the surface or object you want to analyze.
3. Hold the phone steady.
4. Tap the white capture button at the bottom of the screen.
5. Check the image in the gallery before moving on.

For best results, take photos of flat surfaces, windows, doors, walls, floors, vents, or areas where there may be heat loss or temperature differences.

A good example photo would be useful here, especially one showing a clear thermal difference around a window, door, wall, or vent.

Important Camera Settings

MSX Alignment

The FLIR camera combines a visible image from your phone with an infrared image from the FLIR camera. If these images do not line up, use the **MSX Alignment Control** slider to align the actual object and the thermal image.

This is especially important when taking close-up images.

A before-and-after photo of poor MSX alignment and corrected MSX alignment would be useful here.

Emissivity

Use the default **Matte** setting for most non-reflective building materials.

Be careful when taking readings from glass, shiny metal, or reflective surfaces because reflections can affect the temperature reading.

Spot Meter

The spot meter can be used to display a temperature reading on the image. Keep in mind that the reading can be affected by distance, ambient temperature, material type, and reflectivity.

Temperature Units

Use the settings menu to make sure the temperature unit is set to the required unit, such as Celsius.

Calibration

The FLIR camera may freeze briefly while it automatically calibrates. This is normal.

If the image looks inaccurate, blurry, or delayed, try manually calibrating the camera by tapping the calibration icon in the app.

A screenshot showing the calibration icon would be useful here.

Troubleshooting

The camera will not connect to the phone

1. Check that the camera connector matches your phone.
2. Make sure the camera is turned on.
3. Unplug the camera and plug it back in.
4. Close and reopen the FLIR ONE app.
5. Check that the app has permission to use the phone camera.
6. Try restarting the phone.
7. If it still does not work, return the camera to the sign-out station and try another camera.

The camera battery is dead

1. Return to the sign-out station or charging station.
2. Plug the camera into a USB-C charging cable.
3. Use another available camera if needed.

The FLIR ONE Pro takes about 1 hour to fully charge. It is not intended to be charged while in use.

The image is frozen

This may happen during automatic calibration and is usually normal. Wait a few seconds.

If the image stays frozen:

1. Disconnect and reconnect the camera.
2. Close and reopen the app.
3. Turn the camera off and back on.
4. Try manual calibration.

The image does not line up with the object

Use the **MSX Alignment Control** slider to align the visible image and thermal image.

This is most likely to happen when taking close-up photos.

The temperature readings seem incorrect

1. Make sure you are not measuring reflective glass or shiny metal.
2. Use the default **Matte** emissivity setting for most building materials.
3. Move closer to the object if possible.
4. Avoid taking readings through glass.
5. Allow the camera to calibrate.
6. Make sure the spot meter is placed directly on the area being measured.

The image is too dark or lacks detail

The FLIR camera uses visible light to improve image detail. If the area is dark, turn on the phone's flashlight or move to a better-lit area.

The camera gets wet or is used near water

The FLIR ONE and FLIR ONE Pro are not waterproof. Do not use the cameras in wet conditions and return the camera immediately if it may have been exposed to water.

Returning the Camera

1. Return the camera to the sign-out station.
2. Confirm that the camera number matches the one written on the sign-out sheet.
3. Let the event team know if there were any issues with the camera.
4. If the battery is low, place the camera at the charging station.